LSTM vs Transformer

Two sequence encoders compared on AG News 4-class topic classification

Team Jihun Jang (ETRI School · 02521122)

Affiliation University of Science and Technology · ETRI School

Not an accuracy race: we explain WHY the two architectures differ

Task

- **AG News 4-class news topic classification** (World·Sports·Business·Sci-Tech)
- Both LSTM and Transformer Encoder trained from scratch

Method

- **Data, preprocessing, vocabulary, sequence length, training budget** held identical
- Only the encoder differs by architecture

Question

- How do performance, convergence, data efficiency, and error patterns differ?
- And what causes the difference?

Balanced 4-class, vocabulary built from train only (prevents leakage)

Split (seed 42)

- **Train 108,000 / Validation 12,000 / Test 7,600**

Classes

- **World · Sports · Business · Sci-Tech**, balanced
- train ~27,000 / test 1,900 each

Preprocessing (shared by both models)

- word-level tokenization (lowercased)
- vocabulary: **train-only**, max 20,000, min frequency 2
- max length 128, padding positions masked out

Shared backbone, only the encoder mechanism differs

Shared backbone

- **Embedding(128) → Encoder → masked mean pooling → Linear(4)**
- only the encoder is swapped, everything else identical

Core mechanism (basis for interpretation)

- **LSTM** processes tokens sequentially, accumulating context in the hidden state (recurrence)
- **Transformer** connects all tokens directly via self-attention, distributes evidence across tokens

Setup · scale

- **LSTM** bidirectional 2 layers · hidden 128 · params 3,220,484
- **Transformer** 2 layers · 4 heads · FFN 256 · sinusoidal PE · params 2,825,476
- embedding dominates, so the two models are within **~14%** in parameters

Only variable = encoder architecture (everything else fixed)

Fixed (fair comparison)

- split · tokenizer · vocabulary · max length · pooling
- optimizer · learning rate · batch size · epoch · seed

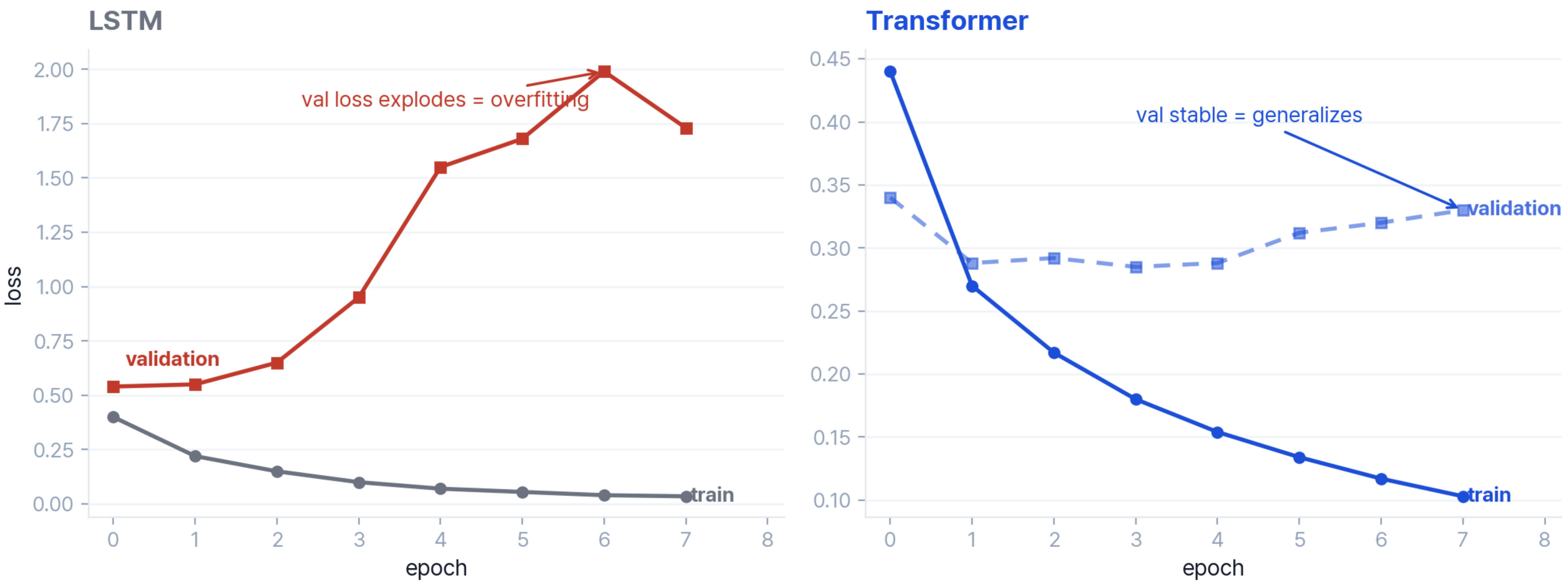
Training

- Adam, learning rate 0.001, batch 64, up to 8 epochs, dropout 0.1, seed 42
- model selection on **validation**, test evaluated once

Ablation · metrics

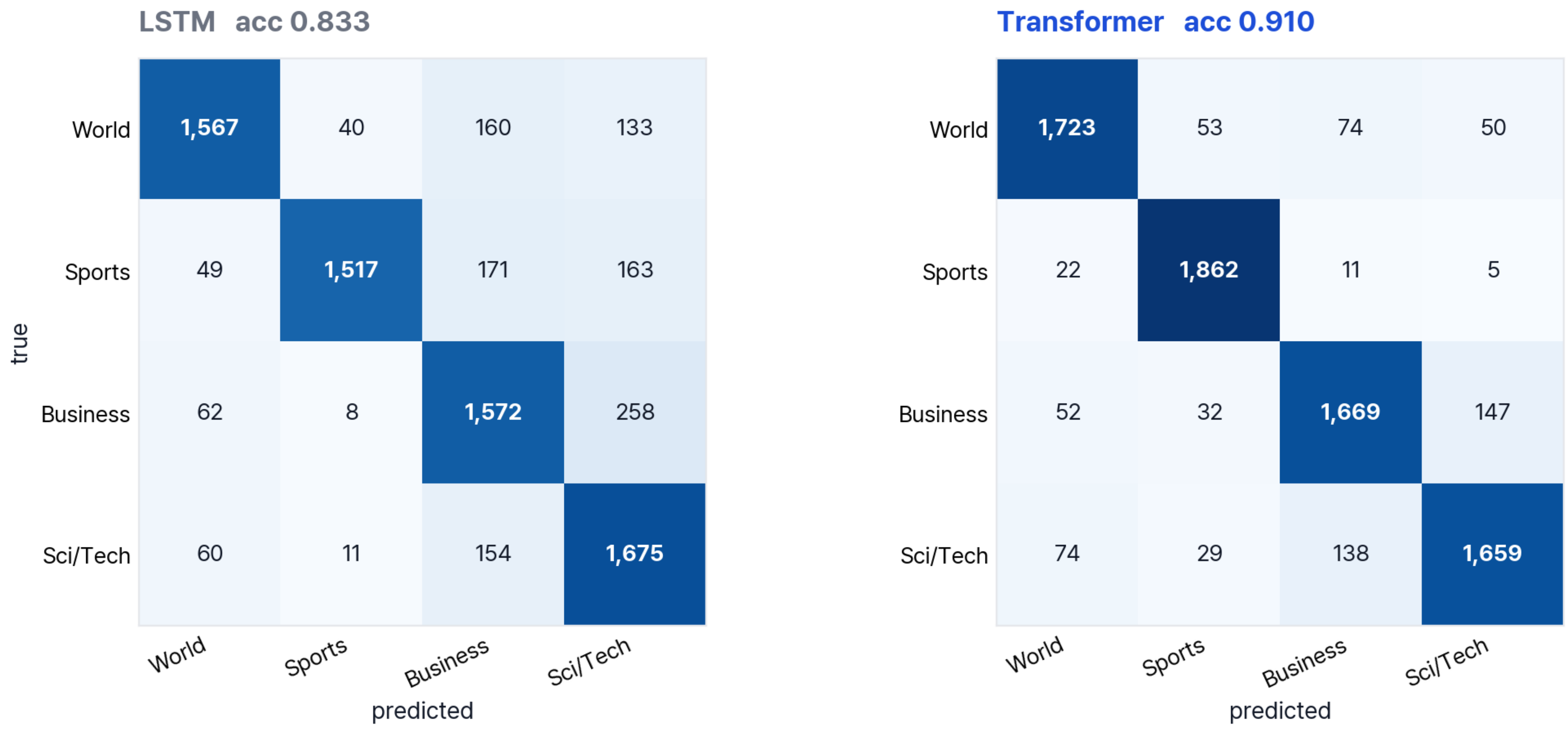
- **training set 5 / 25 / 50 / 100%** (vocabulary fixed, stratified, full val·test)
- accuracy · macro F1-score · loss curve · confusion matrix

Transformer 0.910 vs LSTM 0.833, the gap comes from overfitting



- **Transformer** test accuracy **0.910** · macro F1 0.909
- **LSTM** 0.833 · macro F1 0.835, **val loss explodes**, best epoch 2 saved

Both models confuse Business ↔ Sci/Tech the most



- **Both wrong: 428** 61% on the Business / Sci-Tech boundary (semantic overlap)
- **LSTM-only: 841** spread across all classes, **3.2×** the Transformer-only (259)

LSTM saturates at 25%, Transformer keeps improving

Transformer

- **monotonic increase with data (scaling)**
- more data, better accuracy

LSTM

- **peaks at 25%, then saturation**
- overfitting prevents using more data

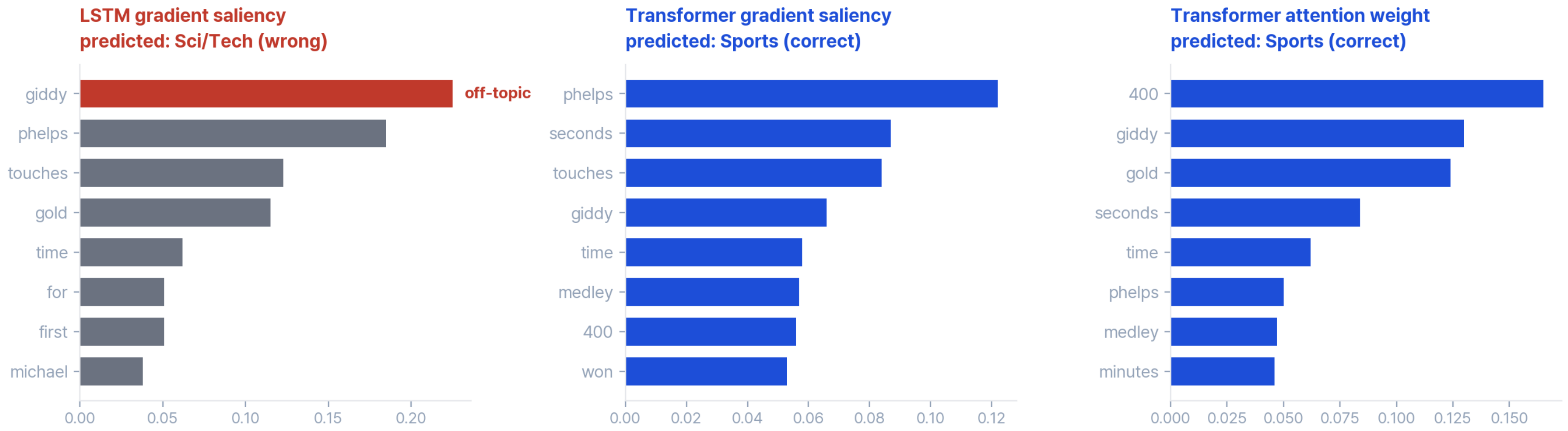
Hypothesis rejected

- 'data-hungry' hypothesis rejected
- Transformer leads across all data sizes

Bag-of-words task: the gap is robustness, not word order

Input (label: **Sports**) "Giddy Phelps Touches Gold for First Time Michael Phelps won the gold medal in the 400 individual medley and set a world record in a time of 4 minutes 8.26 seconds."

Same Sports article: LSTM fixates on one token (wrong); Transformer distributes over topical tokens



- **PE on/off** with-PE 0.910 vs no-PE 0.911 (diff -0.001), word order barely matters
- **robustness** LSTM relies on off-topic 'giddy' and misclassifies; Transformer distributes evidence over topical tokens and classifies correctly

Under identical conditions, Transformer wins, cause is generalization

Performance

- test accuracy **0.910 vs 0.833** under identical conditions (controlled comparison, only the encoder changed)

Cause (mechanism difference)

- **Transformer**: self-attention distributes evidence across tokens → robust, stable scaling
- **LSTM**: recurrence relies on specific tokens → overfitting, saturation at 25%
- the generalization gap between the two mechanisms drives the performance gap

Task nature

- **bag-of-words**: removing positional encoding leaves accuracy unchanged, the gap is robustness not order

Limitations · future

- single seed → confirm with multiple seeds, strengthen LSTM regularization
- re-examine positional encoding on order-sensitive tasks